# PRIMES IS IN P

A presentation on work by Agrawal, Kayal and Saxena

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2026-03-24

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## 1.1 The PRIMES problem

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  - This takes exponential time in the length of input, which is $\log(n)$.
  - We want to do this efficiently (in polynomial time).

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- The algorithm can be made deterministic assuming ERH.

**Theorem (Adleman, Pomerance, and Rumely ‘83).** There is a deterministic $\log(n)^{\mathcal{O}(\log\log\log(n))}$ algorithm for PRIMES.

- The first algorithm that is sub-exponential in time.

## 1.3 This work

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We present a primality testing algorithm that is polynomial time. More exact time-bounds depend on further analysis.

We will prove that the algorithm is $\tilde{\mathcal{O}}\left(\log^{\frac{21}{2}}(n)\right)$ but this can, with some effort, be reduced to $\tilde{\mathcal{O}}\left(\log^{\frac{15}{2}}(n)\right)$.

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Under a widely believed conjecture about the density of Sophie Germain primes (primes $p$ such that $2p + 1$ is also prime), the time complexity is reduced to $\tilde{O}(\log^6(n))$.

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- We denote

$$f(X) = g(X) \pmod{h(X), n}$$

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as $f(X) = g(X)$ in the ring $\mathbb{Z}_n[X]/(h(X))$.

- For any time complexity function $t(n)$, $\tilde{\mathcal{O}}(t(n)) = \mathcal{O}(t(n) \cdot \mathrm{poly}(\log t(n))$
  - Note that for any $\varepsilon > 0$

$$\tilde{\mathcal{O}}(t(n)) = \mathcal{O}(t^{1+\varepsilon}(n))$$

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- Given $r \in \mathbb{N}, a \in \mathbb{Z}$ with $\gcd(a, r) = 1$, the order of $a$ modulo $r$ is denoted as $o_r(a)$. That is, $o_r(a)$ is the smallest $k \in \mathbb{N}$ such that $a^k = 1 \pmod{r}$.

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- Given $r \in \mathbb{N}$, the *Euler totient* $\phi(r)$ is the size of the set $\{a \in \mathbb{N} : a < r, \gcd(a, r) = 1\}$.

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## 2.1 A polynomial primality test

The idea is inspired by the following result

**Lemma.** Let $n \in \mathbb{N}, n \geq 2$ and $a \in \mathbb{Z}$ such that $\gcd(a, n) = 1$. Then,

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$$\binom{n}{q} = \frac{n(n-1)\cdots(n-(q+1))}{q!} = \underbrace{\frac{(n/q)(n-1)\cdots(n-(q+1))}{(q-1)!}}_{\text{not divisible by } q^k}$$

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and $\gcd(a, n) = 1 \implies \gcd(a, q) = 1 \implies \binom{n}{q} a^{n-q} \not\equiv 0 \bmod q^k$, $X^q$ has non-zero coefficient $\binom{n}{q} a^{n-q} \bmod n$, so $(X + a)^n \not\equiv X^n + a \pmod{n}$. ∎

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This lemma gives us a simple test to check primality but is inefficient: we have to compute all $n = 2^{\log n}$ coefficients of these two $n$-degree polynomials.

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The tradeoff is that we need to check this equation for many values of $a$ to get a correct test. We will see, however, that we can still do this in polynomial time.

## 2.2 The AKS Primality Test

**AKS Primality Test**

```
procedure IsPrime(n)
    if n = a^b for a ∈ ℕ, b > 1
        output COMPOSITE
    r ← smallest such that o_r(n) > log^2 n
    if ∃a ≤ r, 1 < gcd(a, n) < r
        output COMPOSITE
    if n ≤ r
        output PRIME
    for a = 1, ..., ⌊√φ(r) log n⌋
        if (X + a)^n ≠ X^n + a (mod X^r − 1, n)
            output COMPOSITE
    output PRIME
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Let $n > 1$. We want to show $\textsc{IsPrime}(n)$ outputs PRIME if and only if $n$ is prime.

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5: **if** $\exists a \le r, 1 < \gcd(a, n) < r$
6: **output** COMPOSITE
7: **if** $n \le r$
8: **output** PRIME
9: **for** $a = 1, \ldots, \left\lfloor \sqrt{\phi(r)} \log n \right\rfloor$
10: **if** $(X + a)^n \neq X^n + a \pmod{X^r - 1, n}$
11: **output** COMPOSITE
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The ($\Longleftarrow$) direction is trivial: when $n$ is prime, the algorithm certainly cannot return COMPOSITE on lines 3 or 6 and the for loop will never identify $n$ as composite by the discussion above.

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The algorithm could not have terminated on line 8, for then the previous step would have found a non-trivial factor. Thus, the algorithm terminates at the end, after the for loop.

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Let $r$ be as chosen in Line 4. Since $o_r(n) > \log^2 n \geq 1$, we have some prime divisor $p \mid n$ with $o_r(p) > 1$. The algorithm did not terminate in lines 6 or 8, so $p > r$.

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Note that we have both $n, p \in \mathbb{Z}_r^*$. Let $\ell = \left\lfloor \sqrt{\phi(r)} \log n \right\rfloor$.

## 3.3 Introspective numbers

**Definition.** For a polynomial $f(X)$ and $m \in \mathbb{N}$, say $m$ is *introspective* for $f(X)$ if

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We claim that $p, n/p$ are introspective for all polynomials $X + a$, where $0 \leq a \leq \ell$.

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The algorithm verifies that for every $0 \le a \le \ell$, we have

$$(X + a)^n = X^n + a \pmod{X^r - 1,\, n}$$

which implies $(X + a)^n = X^n + a \pmod{X^r - 1,\, p}$.

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Putting both equations together, we have

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Since $p \nmid r$, the polynomial $X^r - 1$ is square-free and thus the ring of polynomials mod $X^r - 1$ is reduced. So, we can cancel $p$'th roots and obtain

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Thus, both $p, n/p$ are introspective for all $(X+a)$ where $0 \le a \le \ell$.

## 3.3 Introspective numbers

Our multiplicative closure lemmas then imply that for

$$I = \left\{ \left(\frac{n}{p}\right)^i p^j : i, j \geq 0 \right\} \qquad \text{and} \qquad P = \left\{ \prod_{a=0}^{\ell} (X+a)^{e_a} : \forall a, e_a \geq 0 \right\}$$

every number in $I$ is introspective for every polynomial in $P$.

## 3.3 Introspective numbers

Now consider the following two groups

- $G = \{i \bmod r : i \in I\}$
  - $G$ is generated by $\frac{n}{p}, p$ in modulo $r$
  - This is a subgroup of $\mathbb{Z}_r^*$
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  - This is a subgroup of $\mathbb{Z}_r^*$
  - $|G| > \log^2(n)$ as $o_r(n) > \log^2(n)$.
- $\mathcal{G} = \{f \bmod h(X), p : f \in P\}$ where $h(X)$ is an irreducible factor of degree $o_r(p)$ in $Q_r(X)$, where $Q_r(X)$ is the $r$th cyclotomic polynomial over $\mathbb{F}_p$ that divides $X^r - 1$.
  - This is generated by $X, ..., X + \ell$ in $\mathbb{F} = \mathbb{F}_p[X]/h(X)$.
  - This is a subgroup of the multiplicative group of $\mathbb{F}$.

## 3.4 Size bounds

We'll first establish two facts relating $|\mathcal{G}|$ and $|G|$.

**Lemma.** $|\mathcal{G}| \geq \binom{|G|+\ell}{|G|-1}$.

**Lemma.** If $|\mathcal{G}| > n^{\sqrt{|G|}}$, then $n$ is a power of $p$.

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We will conclude by showing $\binom{|G|+\ell}{|G|-1} > n^{\sqrt{|G|}}$ so that $n$ is a power of $p$. This implies $n = p$ as if $n = p^k$ where $k \geq 2$, this will contradict $\textsc{IsPrime}(n) = \mathsf{PRIME}$ as the algorithm would have found $n = p^k$ in the first step.

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For any $m \in G$, we also have $f(X)^m = g(X)^m$ in $\mathbb{F}$ and thus, as $m$ is introspective for $f, g$, we have $f(X^m) = g(X^m)$ in $\mathbb{F}$.

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So, for all $m \in G$, $X^m$ is a root of the polynomial $f(Y) - g(Y)$ in $\mathbb{F}$. This is a contradiction, as $\deg(f(Y) - g(Y)) < |G|$. So, $f(X) \neq g(X)$ in $\mathbb{F}$.

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*Proof.* We prove the contrapositive: if $n$ is not a power of $p$ then $|\mathcal{G}| \leq n^{\sqrt{|G|}}$.

Let $t = |G|$. Consider the subset $\hat{I} \subseteq I$ defined by

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Since $n$ is not a power of $p$, every distinct pair $(i, j)$ yields a different element, so $|\hat{I}| = \left(\left\lfloor \sqrt{t} \right\rfloor + 1\right)^2 > t$. Since $|G| = t$, there exist $m_1 > m_2 \in \hat{I}$ with $m_1 = m_2 \bmod r$.

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We claim that every element of $P$ is a root of the polynomial $Q(Y) = Y^{m_1} - Y^{m_2}$ in the field $\mathbb{F}$. First, note that from $m_1 = m_2 \bmod r$ we have

$$X^{m_1} = X^{m_2} \pmod{X^r - 1}$$

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So, for any $f(X) \in P$, we have $m_1, m_2 \in \hat{I} \subseteq I$ introspective for $f$ and so

$$(f(X))^{m_1} = f(X^{m_1}) = f(X^{m_2}) = (f(X))^{m_2} \pmod{X^r - 1,\, p}$$

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There are at most $m_1 \leq \left(\frac{n}{p} \cdot p\right)^{\lfloor \sqrt{t} \rfloor} \leq n^{\sqrt{t}}$ such roots so $|\mathcal{G}| \leq |P| \leq n^{\sqrt{t}} = n^{\sqrt{|G|}}$ as needed. $\blacksquare$

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## 4.1 An upper bound on $r$

**Lemma.** Let $\mathrm{LCM}(m)$ denote the lcm of first $m$ numbers. For $m \geq 7$:

$$\mathrm{LCM}(m) \geq 2^m$$

**Lemma.** There exist an $r \leq \max\{3, \lceil \log^5(n) \rceil\}$ such that $o_r(n) > \log^2(n)$.

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*Proof (modified).* $n < 10$ can be checked manually. Assume that $n \geq 10$. Then $B = \lceil \log^5(n) \rceil > 10$. Consider the smallest $r$ that does not divide the product

$$n^{\lfloor \log(B) \rfloor} \cdot \prod_{i=1}^{\lfloor \log^2(n) \rfloor} (n^i - 1)$$

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$$n^{\lfloor \log(B) \rfloor} \cdot \prod_{i=1}^{\lfloor \log^2(n) \rfloor} (n^i - 1)$$

Since $r$ does not divide $n^i - 1$ for all $1 \leq i \leq \lfloor \log^2(n) \rfloor$, $o_r(n) > \log^2(n)$. Hence,

## 4.1 An upper bound on $r$

$$n^{\lfloor \log(B) \rfloor} \cdot \prod_{i=1}^{\lfloor \log^2(n) \rfloor} (n^i - 1) < n^{\lfloor \log(B) \rfloor + \frac{1}{2} \log^2(n)(\log^2(n)+1)} \leq n^{\log^4(n)} = 2^{\log^5(n)} \leq 2^B$$

Now, by minimality of $r$, the numbers $1, \ldots, r - 1$ divide the product.

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Now, by minimality of $r$, the numbers $1, \ldots, r-1$ divide the product. With the above inequality we get $\mathrm{LCM}(r-1) < 2^B$. Combine with the lemma that $\mathrm{LCM}(B) \geq 2^B$, this implies $r \leq B$, as needed. ■

## 4.2 A loose time bound

**Lemma (Joachim von zur Gathen, Jürgen Gerhard '99).** Addition, multiplication, and division operations between two $m$ bits numbers (two degree $d$ polynomials with coefficients at most $m$ bits) can be performed in time $\tilde{O}(m)$ ($\tilde{O}(d \cdot m)$) steps.

## 4.2 A loose time bound

**AKS Primality Test**

**procedure** IsPrime($n$)
  **if** $n = a^b$ for $a \in \mathbb{N}, b > 1$
    **output** COMPOSITE
  $r \leftarrow$ smallest such that $o_r(n) > \log^2 n$
  **if** $\exists a \leq r, 1 < \gcd(a, n) < r$
    **output** COMPOSITE
  **if** $n \leq r$
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  **for** $a = 1, ..., \left\lfloor \sqrt{\phi(r)} \log n \right\rfloor$
    **if** $(X + a)^n \neq X^n + a (\operatorname{mod} X^r - 1, n)$
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Line 4 can be done by trying $O(\log^5(n))$ different $r$'s. Each $r$ needs at most $O(\log^2(n))$ multiplications in modulo $r$, and so will take time $\tilde{O}(\log^2(n)\log(r))$. This entire step takes $\tilde{O}(\log^7(n))$ steps.

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Line 5 computes the gcd of $r$ numbers. Each gcd computation takes time $\mathcal{O}(\log n)$. This entire step takes $\mathcal{O}(r \log(n)) = \mathcal{O}(\log^6(n))$ steps.

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Line 9 verifies $\sqrt{\phi(r)} \log(n)$ equations.

Each equation requires $\mathcal{O}(\log(n))$ multiplications of degree $r$ polynomials with coefficients of size $\mathcal{O}(\log n)$.

So each equation can be verified in $\tilde{\mathcal{O}}(r \log^2(n))$ steps. This entire step takes $\tilde{\mathcal{O}}\left(r\sqrt{\phi(r)} \log^3(n)\right) = \tilde{\mathcal{O}}\left(r^{\frac{3}{2}} \log^3(n)\right) = \tilde{\mathcal{O}}\left(\log^{\frac{21}{2}}(n)\right)$ steps.

Hence, the time complexity of the algorithm is $\tilde{\mathcal{O}}\left(\log^{\frac{21}{2}}(n)\right)$.

## 4.3 Time improvements

**Conjecture.** The number of primes $q \leq m$ such that $2q + 1$ is also a prime is asymptotically $\frac{2C_2 m}{\ln^2(m)}$ where $C_2$ is the twin prime constant (estimated to be approximately 0.66).

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**Remark.** If the above conjecture holds then $r = \mathcal{O}(\log^2(n))$ instead of $\mathcal{O}(\log^5(n))$. This gives a time complexity of $\tilde{\mathcal{O}}(\log^6(n))$.

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**Remark.** If the above conjecture holds then $r = \mathcal{O}(\log^2(n))$ instead of $\mathcal{O}(\log^5(n))$. This gives a time complexity of $\tilde{\mathcal{O}}(\log^6(n))$.

**Lemma. (Baker, Harman '96)** Denote $P(m)$ as the greatest prime divisor of $m$. There exists $n_0$ such that for all $x \geq n_0$,

$$\left|\left\{q \mid q \text{ is prime, } q \leq x \text{ and } P(q-1) > q^{\frac{2}{3}}\right\}\right| \geq 0.6683 \frac{x}{\ln(x)}$$

## 4.3 Time improvements

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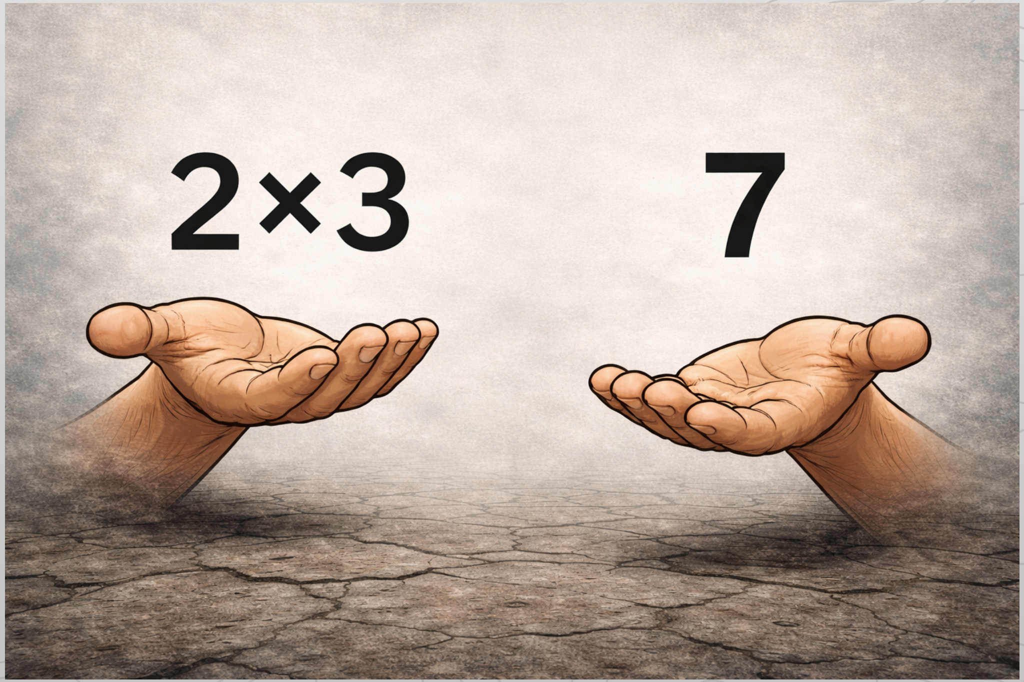
**Remark.** The above lemma shows the algorithm has a time complexity of $\tilde{\mathcal{O}}\left(\log^{\frac{15}{2}}(n)\right)$.

## 4.4 Conclusion

We have presented a polynomial time algorithm that can test whether a number is prime.




Thank you to Professor Shubhangi Saraf for her invaluable insight during the semester and to our classmates for the many illuminating discussions.